Thematic week: Water and Land

Thematic axis: Land Use Planning, Forest Cover and Afforestation

Títle of the presentation: Climate change and the water cycle in Southern Europe: The role of critical thresholds and feedbacks

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Abstract:

Around the Mediterranean sea, deserts and desert-like conditions can be found in close proximity to a very warm sea and thus to a marine airmass with a high moisture content, e.g., the coasts of Morocco, Algiers, Tunisia, Libya, and Almeria in Southeastern Spain. These regions were covered with vegetation in historical times, e.g., during the Roman Empire, and in the case of Almeria, just 150 years ago, before the forests were used to fuel lead mines. The question is: did they run a feedback cycle towards desertification by removing the forests? The results from seventeen European Commission Research Projects (acknowledgements) suggest that this could be the case.

This work shows that the hydrological system in this region is very sensitive to land-use changes and, more recently, to air pollution effects as well. Both of these can combine to exceed critical threshold levels, e.g, the height of the cloud condensation levels with respect to the height of the coastal mountain ranges. This results in the loss of summer storms and tips the regional climate towards desertification and drought. In the Western Mediterranean Basin the non-precipitated water vapour returns and accumulates over the sea, to heights reaching over 5000 m, for periods lasting from 3 to 10 days in summer.

Moreover, changes and perturbations to the hydrological cycle in any part of the basin could propagate to the whole basin and adjacent regions, and ultimately to the global climate system, through other mechanisms. These involve: (1) an increase in Mediterranean cyclogenesis in autumn-winter through cumulative (greenhouse) heating of the sea surface by the water vapour and the pollutants (ozone) accumulated over the sea, (2) the export of the accumulated water vapour and pollutants to other regions at the end of each 3-10 day accumulation-recirculation cycle, which can contribute to summer floods in Central-Eastern Europe and, (3) changes in the evaporation-precipitation balance over the Mediterranean, which increases its salinity and drives the Atlantic-Mediterranean salinity valve.

Key words: Climatic Feedbacks, Drought, Hydrological Cycle, Mediterranean Climate

Climate change and the water cycle in Southern Europe: The role of critical thresholds and feedbacks

1. Introduction

One issue in climate is feedbacks. "Predictions of global atmospheric models are highly sensitive to prescribed large-scale changes in vegetation cover", and "although available studies illustrate the potential effects of massive vegetation changes on the climate system, they can hardly be validated" (Claussen 2001). Nevertheless, the fact that the Western Mediterranean Basin (WMB) is a deep sea surrounded by mountains in the subtropical latitudes makes it an ideal testground for checking whether or not vegetation is a passive component of climate, for determining its role in drought and desertification, and for investigating other questions related to feedbacks in climate studies.

Around the Mediterranean, deserts and desert-like conditions are found in close proximity to a warm sea and, thus, to a marine airmass with a high moisture content, e.g., the coasts of Algiers, Tunisia, Libya, and Almeria in Southeastern Spain. These regions were covered with vegetation in historical times, e.g., during the Roman Empire (Bölle 2003a). In Almeria, dense oak forests covering the mountains were cut down to fuel mines just 150 yr ago (Charco 2002). The question is: did these areas run a feedback cycle towards drought and desertification as a consequence of removing the forests and desiccating the coastal marshes? The experimental data and modelling results from several European research projects suggest that this could be the case.

One result of these projects was the disaggregation of precipitation componentes from: (1) summer storms driven by seabreezes, (2) "classic" Atlantic frontal precipitation and (3) Mediterranean cyclogenesis. All of these, it should be strongly emphasised, respond differently to known climatic indexes, e.g., the NAO (Millán et al. 2005b). Other results show that the hydrological system in the WMB is very sensitive to land-use changes. For example, consider the airmass in a seabreeze. As it moves inland its water vapour content increases by evaporation from the surface at the same time that its potential temperature also rises by sensible heating from the surface. The balance between the heat gained and the moisture accumulated determines the airmass' Cloud Condensation Level (CCL), which will become a critical threshold if forced to rise above the coastal mountains because of lack of moisture. This would inhibit the development of summer storms (Millán et al., 2005a) and tips the local climate towards increasing drought.

The latter situation now prevails in the WMB where the seabreezes, their return flows aloft, and their compensatory subsidences over the sea become self-organized in closed vertical recirculations (Figure 1) that extend to the whole basin from April to early October for periods lasting 3 to 10 consecutive days (Millán et al., 1997; Gangoiti et al., 2001). This situation affects the coasts of Northern Africa, the Iberian peninsula, southern France and southern Italy, and suggests that land use perturbations accumulated over historical time (Bölle 2003b), and accelerated in the last 30 years, may have induced changes from an open monsoon-type rain regime in the past, with frequent summer storms over the coastal mountains, to a current regime dominated by closed vertical recirculations and fewer storms. In the presenr situation the non-precipitated water vapour then follows the return flows of the breezes aloft and accumulates over the sea to heights reaching over 5000 m.

Thus, in contrast with regions dominated by advection, pollutants and water vapour can accumulate over the western Mediterranean sea in layers piled over the sea. And, without

requiring the high evaporation rates of more tropical latitudes, these mechanisms can generate a very large, polluted, moist, and potentially unstable airmass after a few days (Figures 2b, 3, 4a). Finally, the accumulated airmass can be uplifted by a transitory depression, or a trough of cold air aloft, enabling the cycle to start anew. The uplifted airmass can then feed onto a $\mathbf{V}_b$ depression track (Figure 4b) and contribute to intense summer precipitations in Central Europe (Ulbrich et al., 2003). Alternatively, this airmass can be vented along the southern Atlas corridor towards the Atlantic (Figure 5).

Moreover, perturbations to the hydrological cycle in any part of the basin can propagate to the whole Mediterranean basin and adjacent European regions and, ultimately, to the global climate system, through other linked mechanisms: (1) increasing Mediterranean cyclogenesis in autumn (Pastor et al. 2001) through cumulative (greenhouse) heating of the sea caused by the water vapour and ozone accumulated over the sea, (2) exporting "en masse" the accumulated moisture to other regions after each 3-10 day accumulation-recirculation period (Ulbrich et al. 2003; Gangoiti et al. 2006) and, as a result, (3) changing the evaporation-precipitation balance over the Mediterranean, which increases its salinity and drives the Atlantic-Mediterranean salinity valve (Kemp-Shellnhuber 2005).

Finally, Figure 6 presents a hypothetical framework linking Western Mediterranean Basin (WMB)-specific atmospheric-oceanic processes, and their possible feedbacks, to effects at the hemispheric (Ulbrich et al. 2003) and global scales (Hamelin 1989; Savoie et al. 1992; 2002; Prospero and Lamb 2003, Kemp-Shellnhuber 2005; Gangoiti et al. 2006).

The available results and data indicate that these processes are already operating, and suggest that fundamental changes, and long-term perturbations to the European water-cycle, are taking place right now. The questions raised are fundamental for European Union water policies in Southern Europe and neighbouring regions, especially since feedback processes on the hydrological cycle cannot be properly simulated in the Global Climate models used to assess future water scenarios for Europe, or for other regions still dominated by monsoon-type precipitations.

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♦ The first experimental data used for this work were obtained during the European Commission Campaigns on Remote Sensing of Air Pollution in: **LACQ** (France, **1975**), **TURBIGO** (Po Valley, Italy, in 1979) and **FOS-BERRE** (Marseille, France, 1983). Additional experimental and modelling results come from the European Commission research projects: **MECAPIP** (1988-1991), **RECAPMA** (1990-1992), **SECAP** (1992-1995), **T-TRAPEM** (1992-1995), **MEDCAPHOT-TRACE** (1993-1995), **VOTALP I** (1995-1998), **VOTALP II** (1995-1998), **BEMA I** (1993-1995), and **BEMA II** (Phase II/ 1998-2000), **MEDEFLU** (1998-2000), **RECAB** (2000-2003), **ADIOS** (2000-2003), **CARBOMONT** (2001-2004), and **FUMAPEX** (2001-**2005**).

Climate change and the water cycle in Southern Europe: The role of critical thresholds and feedbacks

This work is dedicated to the memory of Dr. Heinrich (Heinz) Ott (†2004) for his initial support of this research in 1985 and for his 1993 request to explain the loss of summer storms around the Mediterranean, and Dr. Anver Ghazi (†2005), for his continued encouragement and support of this line of research.

2. Figures

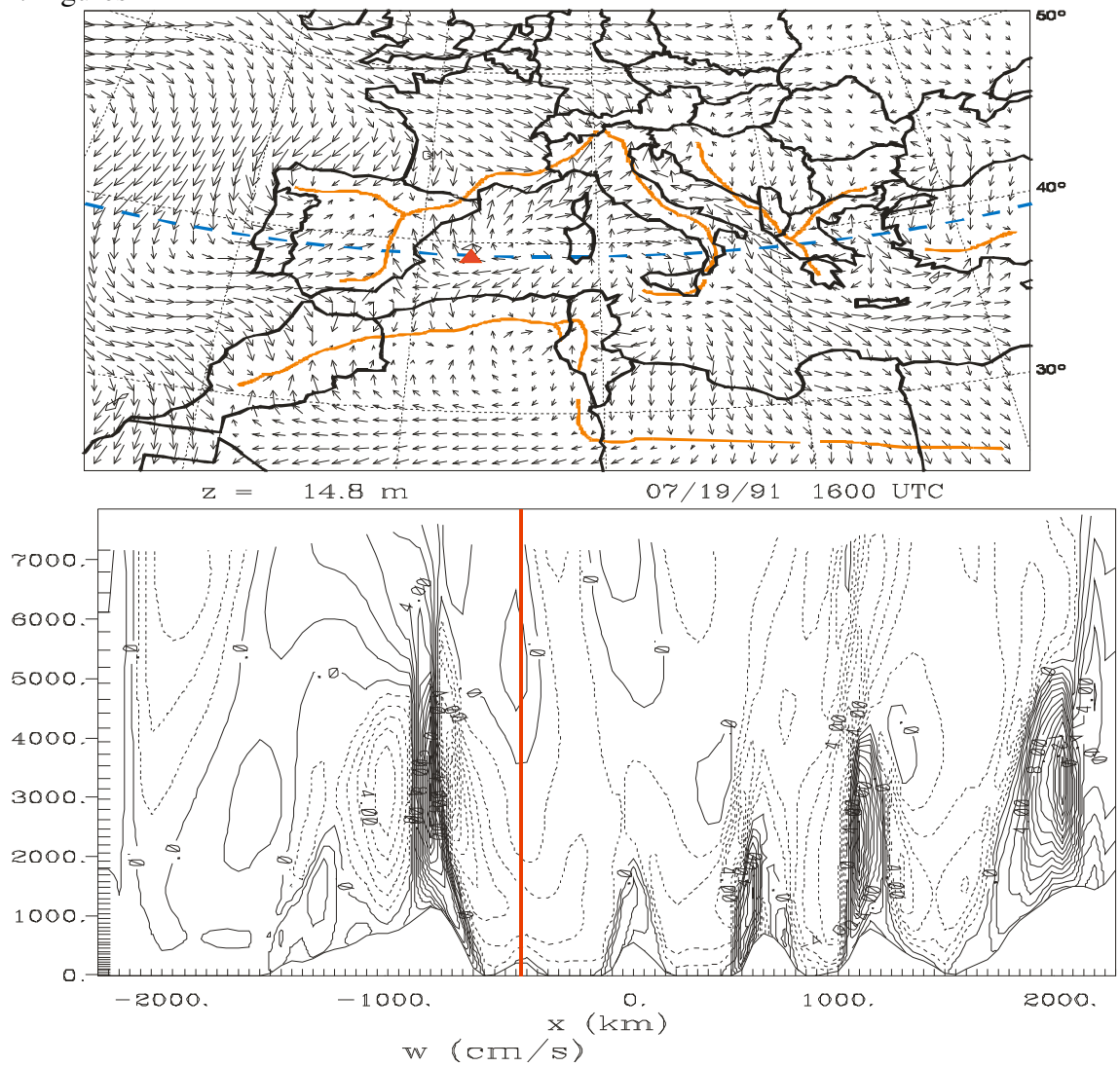
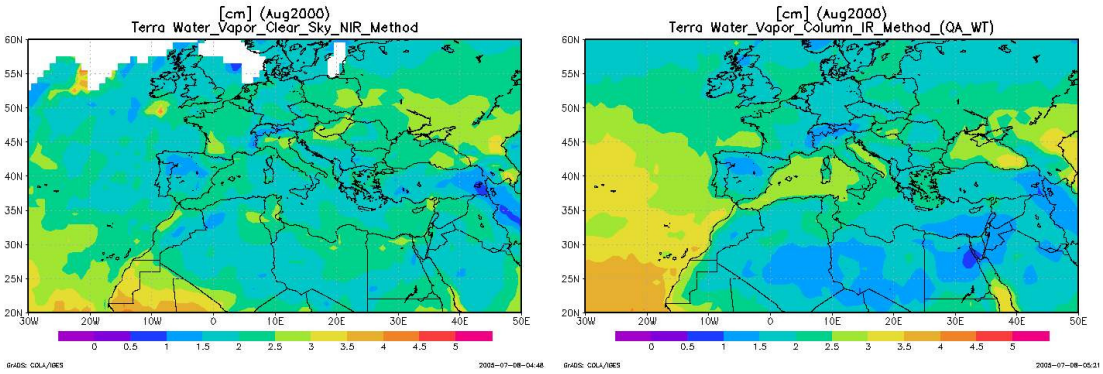
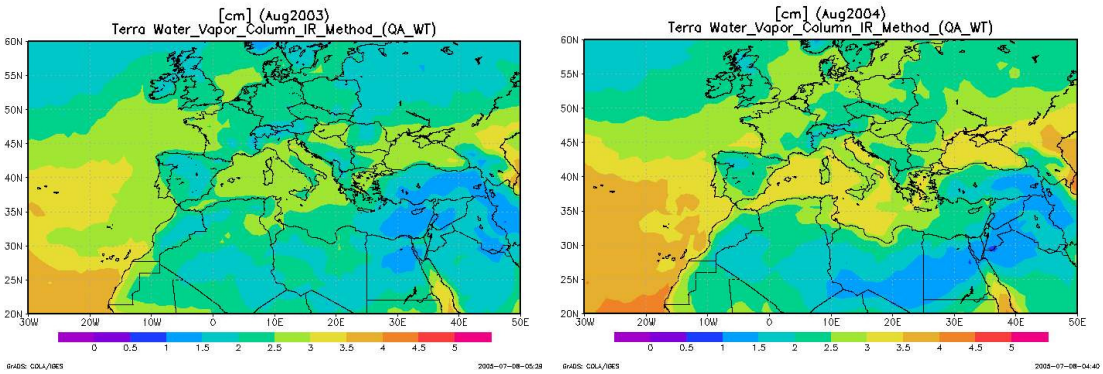


Figure 1. Simulated wind field over the Mediterranean at 16:00 UTC on 19 July 1991, corresponding to a RECAPMA instrumented flight on the same day (15:57 UTC to 16:39 UTC) over the point marked by a red triangle and the vertical line just south of Majorca. **Top graph**: The winds at 14.8 m above the surface emerge from the centre of the western basin and increase in speed while flowing anticyclonically (clockwise) towards the convergence lines located over the mountain ranges surrounding the basin (in orange). **Bottom graph**: The vertical component of the wind speed along the 39.5 North Parallel (dotted blue line in the upper graph) shows deep orographic/convective injections over Eastern Spain and, following to the right, over Sardinia and the west-facing coasts of Italy, Greece and Turkey. Continuity requires compensatory subsidence over the sea to replace the surface air moving towards the coasts, and thus the airmass over the sea sinks (dotted lines). These processes generate a vertical recirculation that piles up layers of pollutants (and water vapour) over the sea to more than 5000 m high. Available measurements suggest that 1/4 to 1/3 of the layers accumulated over the sea during the previous day(s) are recirculated each day. The modelled sinking speeds, however, are much lower than those measured experimentally during the instrumented flight (Millán et al., 2002). The specific structure of the winds is also conditioned by the orientation of the surface with respect to the sun, which at this time of day is 60° West. The time lag required for the circulations to develop and their inertia are also significant, e.g., at this time, upward motions still remain over the west-facing coast of Turkey but are barely developed over Portugal.



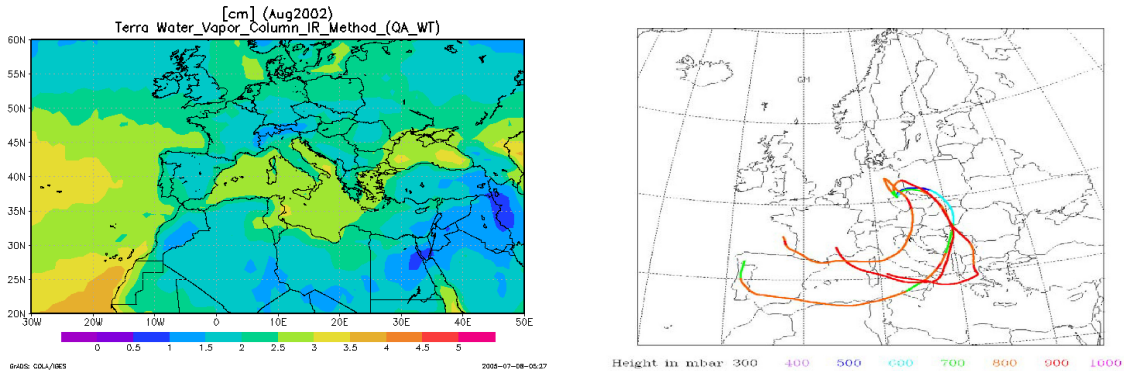
(a) Day **(b) Day + Night**

Figure 2. Averages of the NASA MODIS-Terra measurements (King et al. 2003) for August 2000. The water vapour is considered a tracer of opportunity of the airmasses recirculated by the coastal wind system in the Western Mediterranean Basin. **(a):** The Day product derived from the morning pass at 10:30 UTC emphasises the areas where the satellite looks down the deep orographic-convection developing at the seabreeze fronts over the mountains surrounding the basin (Figure 1), and over desert areas of northern Africa. **(b):** The Day + Night product shows the average of the satellite measurements at 10:30 UTC plus 22:30 UTC, and highlights the areas over which water vapour accumulation occurs, i.e., the WMB, the Adriatic and the Black seas. Accumulation is weaker over the eastern basin, in spite of higher sea surface temperatures and more evaporation, because the atmospheric flows are dominated by advection (Millán et al., 1997). These MODIS products yield the water vapour signal only and eliminate the data in pixels where condensation is detected (cloud masking). This explains some of the low water vapour column values observed over regions where storms develop frequently on summer days (e.g., Alps, Apennines and Atlantic Mid-Atlas in Morocco).



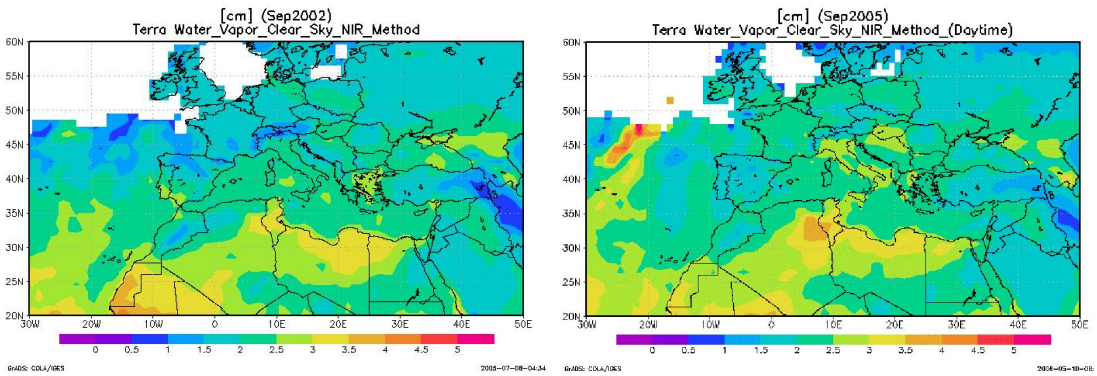
Day + Night **Day + Night**

Figure 3. Monthly averages of the MODIS Day + Night product for August 2003 and 2004. Together with the equivalent products for August 2000 (in Figure 1b) and August 2002 (in Figure 4a), these graphs illustrate the evolution of the average water vapour accumulated over the Mediterranean Basin by the coastal circulations in August for these years. They emphasise the "closed" nature of the coastal circulations at this time, with the result that water vapour accumulates over the sea instead of precipitating over the coastal mountain ranges.



(a)Day + Night

Figure 4. Left. MODIS Day + Night product averaged for August 2002. This value shows the average water vapour accumulated in August 2002 over the western Mediterranean and available for advection to other regions. The graphs at right from Ulbrich et al. (2003) show the back trajectories (type $\mathbf{V_b}$) that fed torrential rains in Germany and the Czech Republic on 11-13 August, 2002. These figures illustrate the evident interconnection between processes from the local to the regional scale in Southern Europe and, possibly, further to the global scale.



(a) Day **(b) Day**

Figure 5. Monthly averages of the MODIS Terra Day product for September 2002, and for September 2005. These graphs support previous working hypotheses on the transport of Mediterranean airmasses along the southern Atlas corridor as one of the outputs of the basin during the vertical recirculation/accumulation periods (Millán et al., 1997). They also illustrate how much water vapour, together with air pollutants, can be transported from the Mediterranean sea to the Atlantic ocean over the Sahara desert. This advected moisture could help in the formation of shallow clouds in any up-slope winds developing on the south-and-east facing slopes of the Atlas mountains. Moreover, these shallow clouds can provide the right environment for heterogeneous reactions involving Saharan dust and pollutants from the Mediterranean area. With respect to drought, the figures also illustrate how a large amount of water vapour (i.e., nearly 3 precipitable cm) can be transported over desert areas without producing rain. This raises two questions: whether past vegetation in these areas could have provided the additional moisture required to trigger precipitation (late summer storms) in former times, and whether the Spanish east coast and other Mediterranean areas are now evolving towards a similar situation by removing vegetation and dessiccating marshes, thereby diminishing the sources of additional moisture required to keep the Cloud Condensation Levels below their critical thresholds.

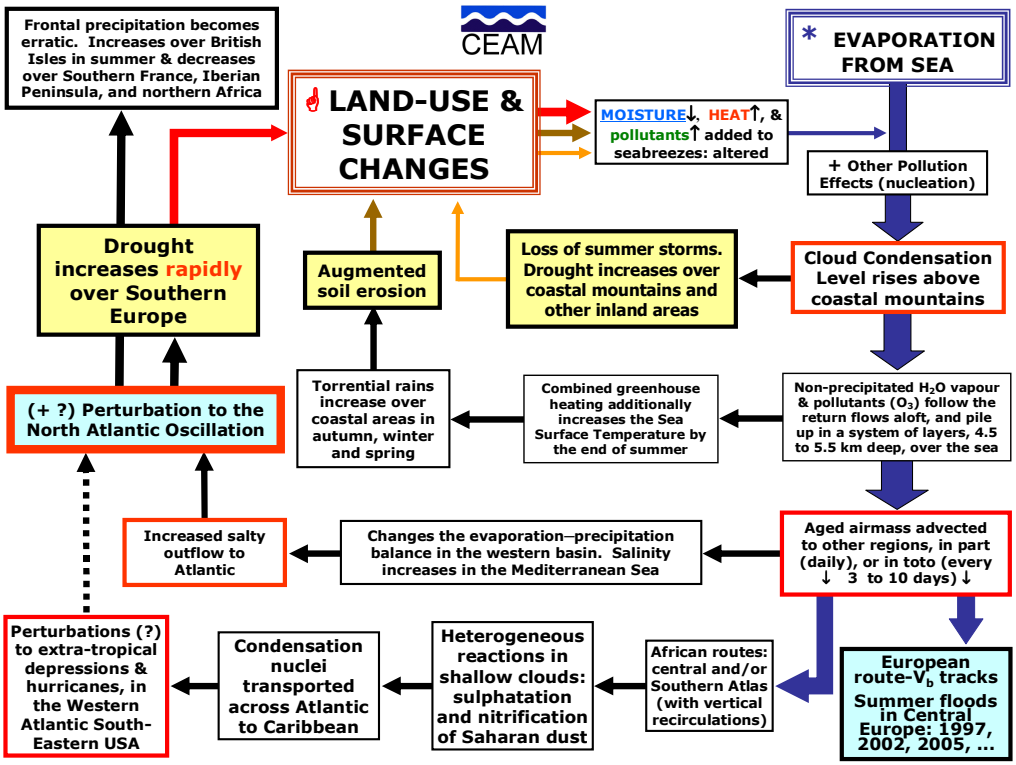


Figure 6. Feedback loops between land-use perturbations in the Western Mediterranean basin and the climatic-hydrological system from the local through the regional to the global scales. The first, local, loop involves the seabreezes and the storms that develop in the afternoon over the coastal mountain ranges. It has a diurnal cycle and a scale of the order of 100 km to 300 km for the surface inflow and the return flows aloft, and it can be repeated for 3 to 10 consecutive days within the western Mediterranean Basin. The regional loop influences the evolution of the Sea Surface Temperature in the western basin during the summer. This warm(er) water then feeds torrential rains in autumn, and more recently also in winter and spring. Finally, the Atlantic-global loop has two components which can affect the North Atlantic Oscillation (NAO): the output of saltier water to the Atlantic and the possible perturbations to the extra-tropical depressions and hurricanes in the Gulf of Mexico generated by changing the characteristics of the Saharan dust transported across the Atlantic. In this figure the path of the water vapour is marked by dark blue arrows, the directly related effects by black arrows, and the indirect effects by other colours. Critical thresholds are squared in red.

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